

AMEYA PANSE

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SUMMARY

AI engineer with 3+ years designing and building LLM-powered systems for cybersecurity and enterprise data problems — spanning threat intelligence pipelines, custom NLP models, knowledge-graph-backed RAG, and multi-agent frameworks deployed on AWS. Full-stack across model development and fine-tuning, retrieval architecture, agentic orchestration with LangChain and LangGraph, FastAPI backends, and cloud-native infrastructure. Domain grounding in automotive cybersecurity — ISO/SAE 21434, TARA, MITRE ATT&CK.

TECHNICAL SKILLS

Domain Expertise: ISO/SAE 21434, UNECE R155, TARA, MITRE ATT&CK, EMB3D, Firmware & Supply-Chain Security

GenAI Engineering: LangChain, LangGraph, OpenAI / Claude / Gemini APIs, Hugging Face, RAG, Multi-agent orchestration, Prompt Engineering

Machine Learning & Data: TensorFlow, PyTorch, pandas, SQL, Regression, Classification, NLP, NER, Conditional Random Fields (CRF), Fine-tuning

Backend Development: FastAPI, RESTful API, Serverless microservices architecture

Cloud & DevOps: AWS services, Docker, Git

Databases: PostgreSQL, MongoDB, Neo4j, FAISS

Languages: Python, C, C++

EXPERIENCE

AI Engineer

Apr 2026 – Present

Equations Work · Pune, India

- Designed and built specialized sub-agents for an enterprise agentic-AI engagement with a global manufacturing client, decomposing and routing complex user queries to the right tools and data sources so the assistant handles multi-step requests reliably.
- Architected the sub-agent orchestration layer — defining each agent's responsibility, hand-off logic, and guardrails — so queries resolve through coordinated specialized agents rather than a single monolithic prompt.
- Built a data pipeline for a manufacturing analytics engagement, processing large volumes of operational data and combining data science with agentic AI to surface actionable insights for the client.

Founding Engineer

Jan 2025 – Mar 2026

Arkcore · Pune, India

- Interacted with clients and internal stakeholders to understand their needs, propose AI-driven architectures, and align on solutions — navigating real constraints like data availability, latency budgets, LLM token costs, and integration complexity to deliver systems that fit the client's actual context.
- Designed and built full-stack, LLM-powered applications on AWS for intelligent data processing, threat analysis, and automated decision-making — owning the work from architecture through deployment.
- Built multi-agent frameworks orchestrating LLMs and automation pipelines to strengthen threat intelligence, compliance mapping, incident correlation, and response.
- Implemented a domain-specific RAG pipeline that improved response accuracy and relevance for cybersecurity knowledge querying.

Associate Software Engineer

Jul 2023 – Dec 2024

SecureThings · Pune, India

- Developed a custom NER model (Conditional Random Fields with domain-specific feature engineering) to extract cybersecurity entities — assets, threats, CVEs, attack vectors — from unstructured security event reports, turning messy incident text into structured, usable data.
- Built a RAG pipeline on Neo4j over an automotive cybersecurity knowledge graph, connecting related threats, assets, and mitigations for richer, relationship-aware retrieval.
- Contributed to an AI-driven ISO/SAE 21434 Compliance Assistant using LLMs to validate evidence, identify compliance gaps, and generate clause-wise cybersecurity reports — producing audit-ready deliverables clients and assessors could rely on.
- Contributed to a TARA Report Gap Analysis Tool that flagged missing threats, assets, and mitigations in risk-assessment workflows — helping ensure assessments were complete and defensible before reaching the OEM.
- Supported client interactions on select project workstreams — presenting outputs, gathering requirements, and coordinating on delivery needs.

Engineering Intern

Jan 2023 – Jun 2023

SecureThings · Pune, India

- Performed cybersecurity data analysis and annotation that became the foundation for downstream models, improving data-handling efficiency by ~30% through optimized workflows and automation.
- Applied NLP techniques — text classification, entity recognition, and predictive modelling — to improve threat-detection accuracy.
- Collaborated on automotive cybersecurity projects integrating compliance requirements with AI-driven attack-path analysis under ISO/SAE 21434.

EDUCATION

Master of Science: Scientific Computing

Jun 2023

Savitribai Phule Pune University, Pune · GPA: 9.45

CERTIFICATIONS

Machine Learning — Electronics & ICT Academy, IIT Roorkee